Inch-Pound

MIL-W-80253B
28 February 1990
SUPERSEDING
MIL-W-80253A
11 February 1985

MILITARY SPECIFICATION

WELDING MACHINES, ELECTRICAL, RESISTANCE, SPOT, PRESS TYPE SINGLE PHASE, BENCH OR FLOOR MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bench or floor mounted, single phase, press type, spot, electrical resistance, welding machines, with equipment and accessories specified herein.

1.2 Classification. The welding machines shall be of the following styles and sizes. The style and size to be furnished shall be as specified (see 6.2).

Style A - Bench mounted.
Style B - Floor mounted.

Size 01 -	3 KVA	transformer rating -	8 inch throat depth.
Size 02 -	5 KVA	transformer rating -	8 inch throat depth.
Size 03 -	10 KVA	transformer rating -	8 inch throat depth.
Size 04 -	15 KVA	transformer rating -	8 inch throat depth.
Size 05 -	20 KVA	transformer rating -	8 inch throat depth.
Size 06 -	30 KVA	transformer rating -	18 inch throat depth.
Size 07 -	35 KVA	transformer rating -	8 inch throat depth.
Size 08 -	50 KVA	transformer rating -	8 inch throat depth.
Size 09 -	75 KVA	transformer rating -	18 inch throat depth.
Size 00 -	100 KVA	transformer rating -	18 inch throat depth.
Size 11 -	150 KVA	transformer rating -	18 inch throat depth.
Size 12 -	200 KVA	transformer rating -	18 inch throat depth.
Size 13 -	250 KVA	transformer rating -	18 inch throat depth.
Size 14 -	300 KVA	transformer rating -	18 inch throat depth.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

AMSC N/A FSC 3432

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1.2.1 Throat depth. The welding machine throat depth is the clear unobstructed distance, measured in the welding plane, from the centerline of the welding electrodes to the first obstruction of the machine frame.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-12332 - Welding, Resistance, Spot, Seam and Projection for Fabricating Assemblies of Low-Carbon Steel.

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS) 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

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29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following document(S) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 5 - Resistance Welding Control. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

RESISTANCE WELDER MANUFACTURERS ASSOCIATION (RWMA)

RWMA Bulletin Number 16 - Resistance Welding Equipment Standards. (DOD adopted)

(Application for copies should be addressed to the Resistance Welder Manufacturers Association, 1900 Arch Street, Philadelphia, PA 19103-1403.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete welding machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The welding machine shall be new (not a prototype) and one of the manufacturer's current production models capable of welding with the electrodes mounted in a vertical plane in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The welding machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the welding machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The welding machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical and pneumatic operating controls shall be located convenient to the operator's work station.

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the welding machine, whether or not specifically required herein, shall conform with all requirements of 29 CFR 1910. Additional safety and health requirements shall be as specified in detail (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial welding machine supplied to the commercial market shall be provided for the point of operation and all other parts of the welding machine that present safety hazards. Additional guarding dependent on workpiece size and configuration shall be provided when specified and fully described by the procuring activity (see 6.2).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

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3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine, and its components materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine. All replacement parts shall be interchangeable with their respective original parts, and fit/function with mating components.

3.3 Construction. The welding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion, or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the welding machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

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3.3.7 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(S) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The welding machine shall consist of not less than the following basic components: machine frame and base, enclosure, head, ram, upper arm assembly, lower arm and horn, electrical system, pneumatic system, and water cooling system.

3.4.1 Machine frame and base. The frame and base assembly shall be a steel weldment of box construction, ribbed for maximum strength and rigidity to maintain original alignment and to provide maximum resistance to mechanical deflection. The style "A" machine shall be designed for bench mounting. The style "B" machine shall provide for floor mounting on a steel base or by use of an integral frame. Access to the interior shall be provided to permit inspection and servicing of the equipment installed within the frame. The front frame shall be designed to permit vertical adjustment of the lower knee and horn assembly.

3.4.2 Enclosure. The metal enclosure shall be corrosion resistant material or treated to resist corrosion. Ventilation and cooling shall be adequate to prevent temperature in excess of design requirement, under normal operating conditions of rated power output.

3.4.3 Head, ram, and upper arm assembly. The head, ram, and upper arm assembly shall be designed and constructed to maintain original alignment and to minimize deflection. The guiding mechanism for the ram shall be of the anti-friction type, and shall be protected against dust and foriegn matter. The ram movement shall be actuated by a foot switch, and have fast acceleration and low enertia. The electrode force shall be calibrated and adjustable. Weld current shall flow only after completion of squeeze time. The throat depth, ram stroke, and electrode force shall be as shown in table I.

3.4.4 Lower arm and horn. The lower arm shall be designed and mounted on the frame to have a vertical adjustment of not less than 4 inches. The mounting of the arm shall be such that the alignment of the electrode will not be changed when the arm is adjusted vertically. The arm shall have adjustments to permit motion of the horn and alignment of the lower electrode holder. The horn shall be made of solid copper, and have a diameter as specified in table I. The horn shall have a clamp that will not mar the electrode holder after repeated adjustments. The lower horn in and out adjustment shall be not less than 2 inches.

3.4.4.1 Electrodes and holders. The electrodes and holders shall conform to RWMA Bulletin Number 16. The quantity, size, type, and alloy class of electrodes and holders required shall be as specified (see 6.2).

3.4.5 Electrical system. The electrical system of the welding machine shall include the transformer, timers, meters, suitable circuit protection, disconnect switches, and all controls as required for installation and

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TABLE I. Machine capacity and performance requirements.

Machine size	Welding transformer power rating (KVA) at 50% duty cycle	Throat depth (inches	Short circuit current + 10% (amperes) at 3 inches horn spacing	Electrode force 1/ pounds	Horn diameter (inches)	Standard ram stroke including adjustment (inches)	Weld capacity two pieces AISI 1020 Steel (Mfg. Std. Gage)
01	3	8	5,000	25-300	1.50	1	25
02	5	8	5,000	25-300	1.50	1	25
03	10	8	7,500	25-250	1.25	1	25
04	15	8	11,000	25-550	1.50	1	20
05	20	8	14,000	50-550	1.50	2	18
06	30	18	20,800	2,250	2.50	2	16
07	35	8	32,400	1,000	2.00	2	16
08	50	8	43,000	1,000	2.00	2	16
09	75	18	31,800	2,250	2.50	4	NA
00	100	18	40,800	2,250	3.00	4	NA
11	150	18	48,900	4,000	3.00	4	NA
12	200	18	54,000	9,100	3.00	4	NA
13	250	18	60,000	9,100	3.00	4	NA
14	300	18	65,500	9,100	3.00	4	NA

Requirements are "not less than those shown", where a range is shown, the requirements are from the smaller figure or less to the larger or more.

1/ The pneumatic system being supplied with an air line supply pressure of 80 psi.

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Operation. The electrical system and components shall conform to the requirements of NFPA 70 where applicable, and not in conflict with the requirements of this specification. Each sub-panel shall be arranged so that it may be removed independently of any other sub-panel. All terminals and units shall be clearly identifiable from the machine electrical wiring diagram. Adequate ventilation of the control cabinets shall be provided. Where ventilation is provided by forced air, the air inlet shall be provided with replaceable filters.

3.4.5.1 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.5.2 Input electrical power supply. The machine shall be designed to operate on a single phase, 60 Hz power supply at either 230 or 460 volts. Unless otherwise specified (see 6.2), the machine shall be furnished to operate on 230 volts.

3.4.5.3 Main disconnect switch. The machine shall have a main disconnect switch or circuit breaker. Unless otherwise specified (see 6.2), a circuit breaker shall be furnished and shall be mounted on the machine panel. The main disconnect switch or circuit breaker shall provide a safety interlock when access doors are open to areas that are a hazard to the operator.

3.4.5.4 Welding transformer. The transformer shall conform to the standard of RWMA Bulletin Number 16. The insulation of the transformer shall be not less than the requirements of class B. The transformer shall have taps connected to a selector switch to provide the necessary heat range for welding. When specified (see 6.2), a series-parallel switch shall be provided for additional heat control.

3.4.5.5 Electrical control unit. The electrical control unit shall consist of the welding power control, timing and sequence controls, welding contactor, current adjustment control, auxiliary, and accessory controls. All calibrations, phase shift controls, and similar characteristics, shall be clearly marked in standard terms such as: cycles (cy.), Impulses (imp.), Percent of maximum (% max.).

3.4.5.5.1 Operator control panel. The operator control panel shall have an on-off (weld/no weld) switch, a power "on" light, a current adjustment control, a weld timer selector, auxiliary, accessory control function switches and any other controls or switches required to meet the requirements specified herein. When furnished, all meters shall have an accuracy of $\pm$ 2 percent of full scale reading.

3.4.5.5.2 Welding power control unit. Unless otherwise specified (see 6.2), the machine shall be furnished with a solid state synchronous precision combination welding power control unit conforming to or equivalent to NEMA ICS 5, type S2H. The control unit may be located in a separate sheet metal cabinet, or mounted as a integral part of the machine.

3.4.5.5.3 Timing and sequencing controls. Unless otherwise specified (see 6.2), the machine shall be furnished with a timer and sequencing control

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conforming to or equivalent to NEMA ICS 5, type 7B. The timing control dial shall be calibrated in cycles (cy.) based on the electrical service line frequency of 60 Hz. The timer shall have at least four functions: squeeze, weld, hold, and off. The various timing and sequencing functions shall be grouped on a plug in chassis. Provisions may be made for the removal or addition of extra functions. The control chassis and components shall be mounted in such manner as to prevent damage to the mounted unit during the service life of the machine, and to allow maximum ease of maintenance and replacement. The timing and sequencing functions shall be selected and adjusted by means of knobs or thumb switches mounted on the front or side of the machine.

3.4.5.5.4 Welding contactor. The welding contactor shall be a silicon controlled rectifier (SCR) type. The SCR shall be mounted in a heat sink, and be thermostatically protected. If additional cooling is required to maintain the desired temperature level in the larger size machine, the heat sink shall be forced air or water cooled. When forced air cooling is furnished, the air inlet shall be equipped with a filter and means shall be provided to prevent welding if the air supply is off or restricted. When water cooling is furnished, means shall be provided to prevent welding if the water is shut off or if the supply is restricted.

3.4.5.5.5 Current adjustment control. Unless otherwise specified (see 6.2), continuously variable adjustment of the welding current, from the upper and lower limits of the machine capacity, shall be provided. The adjustment shall be by a potentiometer type control, transformer taps, digital switch selection in increments of 1, or a combination of any of the above. The means of current adjustment provided shall be such that, at the maximum setting of the adjustment dial, the short-circuit secondary current produced shall conform to the requirements specified in table I.

3.4.5.5.6 Control circuit voltage. The control circuit shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage of no greater than 24 volts. Control equipment shall be capable of operating on control circuit voltage which varies as much as ± 10 percent from the voltage for which the equipment is adjusted.

3.4.5.5.7 Auxiliary control functions. When required, auxiliary control functions shall be furnished as specified (see 6.2).

3.4.5.5.8 Accessory controls. Accessory controls shall consist of at least the following: Foot switch, tip dress control, contact pressure gauge switch, electrode force interlock circuit, weld/no weld switch, and repeat on/off switch.

3.4.5.5.8.1 Foot switch. The foot switch shall control the upper electrode assembly and its associated pneumatic system and initiate the welding cycle. When specified (see 6.2), a two-stage foot switch shall be supplied which shall initiate the electrode force in the first stage, and initiate the welding sequence in the second stage. When specified (see 6.2), the machine shall be furnished with dual palm button controls with anti-tie down features.

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3.4.4.5.8.2 Tip dress control. The tip dress control shall operate to lower the upper electrode under a reduced force for cleaning and aligning the electrodes

3.4.5.5.8.3 Contact pressure gauge switch. The contact pressure gauge switch shall prevent the initiation of the welding cycle until the predetermined electrode force has been established.

3.4.5.5.8.4 Electrode force interlock circuit. The electrode force interlock circuit shall prevent the upper electrode from being raised during the welding cycle.

3.4.5.5.8.5 Weld/no-weld (on/off) switch. A weld/no weld switch shall be provided. In the no weld position, the welding sequence will actuate except that no welding current shall flow.

3.4.5.5.8.6 Repeat on/off (repeat/non-repeat) switch. A repeat on/off switch shall be provided. In the off (non-repeat) position only one welding cycle will result with the actuation of the switch. In the on (repeat) position the welding cycle will repeat automatically after a controlled time interval (off time) as long as the switch remains closed.

3.4.6 Pneumatic system. The welding machine shall have a pneumatic system conforming to RWMA Bulletin Number 16, and the following requirements to control the movement of the upper arm assembly and regulate the pressure applied to the work pieces by the electrodes during the welding cycle.

3.4.6.1 Pneumatic system operation. The pneumatic system air cylinder shall be a double action type mounted above the ram and shall operate with not less than 80 PSI input air service. The inlet connection shall be marked for the pressure and flow requirements. The welding head shall operate from a pneumatic system which shall include valving, piping, regulators, gauges, lubricators, moisture drain provisions, wiring, switches, and all related equipment necessary to permit use of any of the electrode force modes, as follows: Constant Low, Constant High, or Variable. Air cushions or air speed valves shall be provided to insure shock free operation. A release means, for lowering the upper electrode under a reduced force for cleaning and aligning the electrodes, shall be provided. The system shall be controlled by a solenoid valve actuated by either a hand or foot switch.

3.4.6.2 Air gauges. Air gauges shall have a face diameter of not less than 4-1/2 inches, and shall be calibrated in pounds per square inch (PSI). Gauge accuracy shall be within ± 0.5% of the maximum reading of the gauge. Means of shock protection shall be provided to prevent gauge damage. The gauge shall be mounted to be visible to the operator.

3.4.6.3 Air regulators. The regulators shall be self-bleeding. Regulators and solenoid valves shall be installed with unions. The regulators shall maintain the pressure at ± 1/2 PSI within a 2-100 range, as long as the supply pressure is 2 PSI higher than the preset regulator pressure. The air cylinder for the upper electrode shall produce the electrode force as specified in table I.

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3.4.7 Water cooling system. The machine shall have a water cooling system with a minimum of one circuit to the electrodes. The cooling system, when supplied with water in accordance with the recommendations of the manufacturer , will operate to prevent the temperature rise during operation from exceeding the limits specified in RWMA Bulletin Number 16. The system shall operate from a shop tap water supply and shall have a single inlet connection into a manifold. The inlet connection shall be marked for pressure and flow requirements. When specified (see 6.2), the system shall have parallel flows from the manifold to the welding transformer, to the secondary winding and circuit conductors, and to the primary winding welding circuit contactors. The system shall have flow to the upper and lower electrodes. The piping in the system shall be corrosion-resistant to tap water service. The system shall have a single outlet, spilling into an open cup drain for a visual check on the water flow. The water flow lines to the electrodes shall have solenoid valves, installed with pipe union connections, to prevent water flow while changing the electrodes. When specified (see 6.2), the system shall have water saving devices, an inlet strainer, and a closed manifold drain system with flow indicators.

3.4.8 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built in lighting system is provided, the adjustable light described above shall not be required.

3.4.9 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the welding machine. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the welding machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the welding machine shall meet the requirements of table I for the welding machine specified. Machine size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.7 Performance. The welding machine shall be capable of producing spot welds conforming to the requirements of MIL-W-12332 under production conditions and all additional requirements of table I and this specification.

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3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial welding machine shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on the machine including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant

3.11.1 Lubrication plate or chart. When specified (see 6.2) a lubrication plate or chart shall be attached to each welding machine. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2) the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each welding machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below and the information required in RWMA Bulletin Number 16, Name and Rating Plates. If the welding machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

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3.12 Workmanship. Workmanship of the welding machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The welding machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each welding machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and

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the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. An acceptance test shall be performed on each welding machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the welding machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each welding machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 degrees to 104 degrees Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. Each welding machine shall be operated at no load for not less than 15 minutes. Proper operation of all controls, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Welding test. After the test in 4.8.2 has been completed and the machine is at normal operating temperature, spot welds shall be made using two equal thicknesses of AISI 1020 mild steel of the maximum gage specified in table I. Thirty spot welds shall be made, spaced at a distance for which no increase in welding current is necessary to compensate for the shunted current effect of adjacent welds. The test welds shall pass the applicable test requirements specified in MIL-W-12332.

4.8.4 Secondary current test. The short circuit secondary current of the machine shall be in accordance with the procedure of RWMA Bulletin Number 16. The current shall conform to the requirements specified in table I.

4.8.5 Electrode force test. The minimum and maximum electrode force of the machine shall be measured, using an electrode force gauge. The measured minimum shall not be more than, and the measured maximum shall be not less than, the values specified in table I.

4.8.6 Rated capacity test. The machine shall be adjusted to the manufacturer's recommended setting to produce its rated welding current at 50 percent duty cycle and operated for 30 minutes of simulated welds. The temperature limits specified in RWMA Bulletin Number 16 shall mot be exceeded. There shall be no overheating, arcing, or malfunctions of the machine.

4.8.7 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.8 shall be as specified and fully described (see 6.2).

4.8.8 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.7 using the procedures given in MIL-STD-462.

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4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-W-45562.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The equipment covered by this specification is intended for use in contractor's plants and Military and Government repair facilities where it is desirable to use single phase equipment for welding and where general repairs and fabrication of sheet metal and wire is required.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Style and size required (see 1.2 and table I).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Specify system of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5.).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.7).

j. Type, size, quantity, and alloy class of electrodes and holders, if required (see 3.4.4.1).

k. Electrical requirements and input power supply , if different (see 3.4.5.2).

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l. Circuit breaker and its mounting, if different (see 3.4.5.3).

m. Welding transformer, if different (see 3.4.5.4).

n. Welding power control unit, if different (see 3.4.5.5.2).

o. Timing and sequence controls, if different (see 3.4.5.5.3).

p. Current adjustment control, if different (see 3.4.5.5.5).

q. Auxiliary control, if required (see 3.4.5.5.7).

r. Foot switch, if different (see 3.4.5.5.8.1),

s. Dual palm bottons ,if required (see 3.4.5.5.8.1)

t. Water cooling system, if different (see 3.4.7).

u. Water saving devices, if required (see 3.4.7).

v. Worklight, if required (see 3.4.8).

w. Hour meter, if required (see 3.4.9).

x. Size and capacity, if different (see 3.5).

y. Specify physical size limitation, if required (see 3.5.1).

z. Additional performance features, if required (see 3.7.1).

aa. Specify and fully describe optional equipment, if required (see 3.9).

bb. Specify applicable repair parts, if required (see 3.10).

cc. Lubrication plate, if required (see 3.11.1).

dd. Specify information to be provided on lubrication plate if different (see 3.11.1).

ee. Nameplate, if different (see 3.11.2).

ff. Inspection conditions, if different (see 4.3).

gg. First article inspection, if different (see 4.4).

hh. Quality conformance inspection, if different (4.5).

ii. Acceptance test, if different (see 4.6).

jj. Additional performance tests, if required (see 4.8.7).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the remarks section of the DD Form 1423 that all technical data furnished should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(S) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

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6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

MIL-W-80253A	MIL-W-80253B
Machine size	Machine size
1 - Deleted	01 - New
2 - Deleted	02 - New
3 - Deleted	03 - New
4 - Deleted	04 - New
5 - Deleted	05 - New
6 - Deleted	06 - New
7 - Deleted	07 - New
8 - Deleted	08 - New
9 - Deleted	09 - New
10 - Deleted	00 - New
	11 - New
	12 - New
	13 - New
	14 - New

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

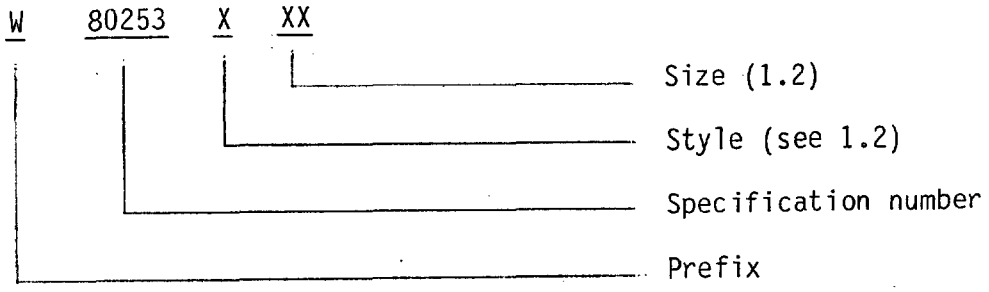
6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PIN to be used for welding machines acquired to this specification are created as follows:

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6.14 Subject term (key word) listing.

Contact pressure gauge switch
Current adjustment control
Electrode force interlock circuit
Lower arm and horn
Pneumatic system
Silicon controlled rectifier
Tip dress control

Custodians:
Army - AL
Air Force - 99

Review activities:
Air Force - 84

User activities:
Army - AR
Navy - AS

Preparing activity:
DLA-IP
Project (3432-0142)

*U.S. GOVERNMENT PRINTING OFFICE: 1990-704-034/30400

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-W-80253B	2. DOCUMENT DATE (YYMMDD) 90 02 28

3. DOCUMENT TITLE: WELDING MACHINES, ELECTRICAL, RESISTANCE, SPOT, PRESS TYPE, SINGLE PHASE, BENCH OR FLOOR MOUNTED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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